THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 7: **Measurement of Time and Motion** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A boat sails 24 km in 3 h with a steady speed. In 5 h it travels: 1

(a) 15 km **(b)** 8 km **(c)** 120 km **(d)** 40 km

2. Timing 20 oscillations in 40.0 s reduces the percentage error to about: 1

(a) 0.25 per cent **(b)** 2.5 per cent **(c)** 5 per cent **(d)** 50 per cent

3. A car goes uphill at 20 km/h and downhill at 60 km/h over the same distance. The average speed is: 1

(a) 40 km/h **(b)** 30 km/h **(c)** 20 km/h **(d)** 80 km/h

4. A child on a merry-go-round has a speed of 3 m/s at a radius of 4 m. Which statement is correct? 1

(a) the speed is constant but the direction changes
(b) it moves along a straight line
(c) speed changes
(d) it is at rest

5. A stopwatch that shows readings up to 0.01 s means: 1

(a) it can measure only 0.01 s **(b)** it runs 0.01 s slow
(c) it measures distance in 0.01 m **(d)** the smallest time it can show is 0.01 s

6. A sprinter takes 10 s for 100 m and a cheetah 4 s for 100 m. The ratio of speeds sprinter : cheetah is: 1

(a) 1 : 1 **(b)** 5 : 2 **(c)** 4 : 10 **(d)** 2 : 5

7. A graph of distance (y) against time (x) is a straight line through the origin with slope 2 m/s. At t = 6 s, distance = 1

(a) 8 m **(b)** 2 m **(c)** 3 m **(d)** 12 m

8. Which of the following is the fundamental reason for using a pendulum as a clock? 1

(a) it is shiny **(b)** equal-time swings
(c) it is long **(d)** it is heavy

9. **Assertion (A):** A car that travels 90 km in 2 hours has a speed of 45 km/h. 1
Reason (R): Speed is equal to distance multiplied by time.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** A speed of 36 km/h is the same as 36 m/s. 1

Reason (R): 1 km/h is equal to $\frac{5}{18}$ m/s.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A bike moves 30 km/h for 20 minutes. Find the distance. 2

12. A swing makes 15 oscillations in 45 s. Find the time period and the number of oscillations in 3 minutes. 2

13. Two runners A and B run 100 m in 12.5 s and 10 s. How far behind is A when B finishes? 2

14. How many seconds in a day? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A car of speed 54 km/h and a bike of speed 12 m/s start together. Who goes farther in 5 minutes and by how much? 3

16. A boy cycles 6 km in 20 min and 9 km in 30 min. Find the average speed and say if the motion is uniform. 3

17. Explain why a bus is not said to move with uniform speed during a journey. 3

OR

The distance-time graph of an object has slope 3 m/s for 4 s, then horizontal for 2 s, then slope 6 m/s for 2 s. Find total distance and average speed. 3

18. Why is a stopwatch always started and stopped by the same person in an experiment? 3

OR

A man walks 1 km in 15 min and then 2 km in 30 min. Find average speed in km/h. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A train from Nagpur travels 144 km in 2 hours at a steady speed. A student, Imran, is asked some questions about its motion. 5

(i) Find the speed of the train in km/h.

(ii) Convert this speed into m/s.

(iii) How far will it go in 3.5 hours at the same speed?

(iv) How much time will it take to cover 360 km at the same speed? State the assumption you make.

20. A cyclist rides 20 km at 10 km/h, then 30 km at 15 km/h. Find the total time and average speed. Compare with the simple average of the speeds. 5

OR

A satellite takes 90 minutes to circle the Earth. Find the number of orbits in a day and the time period in seconds. 5

END OF QUESTION PAPER — Total: 40 Marks